

PHQ-9/GAD-7 SCORE DISTRIBUTION ANALYSIS

(ANA-2)

Dataset Used: Global Mental Health Dataset 2025
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Project: AI Mental Health Assistant

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1. INTRODUCTION

Mental health analytics plays an important role in understanding the distribution and severity of psychological conditions across different demographic groups. Depression and anxiety are among the most commonly reported mental health concerns globally, and analysing their severity patterns can support better interpretation of population-level mental health trends.

This analysis focuses on:

- depression severity indicators,
- anxiety severity indicators,
- demographic segmentation by gender and age,
- and statistical visualization of score distributions.

The task was completed using a structured mental health survey dataset containing demographic information and mental health severity scores. Box plots and violin plots were used to analyse the spread, variability, and distribution density of depression and anxiety scores across demographic groups.

2. OBJECTIVE

The objective of this analysis was to:

- analyse depression score distributions across gender and age groups,
- analyse anxiety score distributions across gender and age groups,
- identify score variability and outliers,
- study demographic distribution patterns,
- and generate comparative statistical visualizations.

3. DATASET OVERVIEW

The analysis was performed using an open structured mental health dataset containing:

- demographic attributes,
- depression severity indicators,
- anxiety severity indicators,
- and related mental health variables.

Key Variables Used:

Variable	Description
Gender	Participant gender category
Age	Participant age
Depression_Score	Depression severity indicator
Anxiety_Score	Anxiety severity indicator

The dataset was cleaned and filtered before analysis to remove missing values and retain only the required analytical variables.

4. TOOLS & TECHNOLOGIES USED:

Tool / Technology	Purpose
Python	Data analysis workflow
Pandas	Data preprocessing and cleaning
Matplotlib	Chart generation
Seaborn	Statistical visualization
Jupyter Notebook	Analysis execution
GitHub	Version control and submission

5. DATA PREPROCESSING:

The preprocessing stage included:

- loading the dataset,
- checking column structure,
- filtering required variables,
- and creating grouped age categories for demographic analysis.

Age values were grouped into:

- 0–18
- 19–30
- 31–45
- 46–60
- 60+

This grouping improved demographic comparison and visualization readability.

6. GENDER-BASED PHQ/GAD DISTRIBUTION ANALYSIS

6.1 PHQ-9 Equivalent Depression Score Analysis:

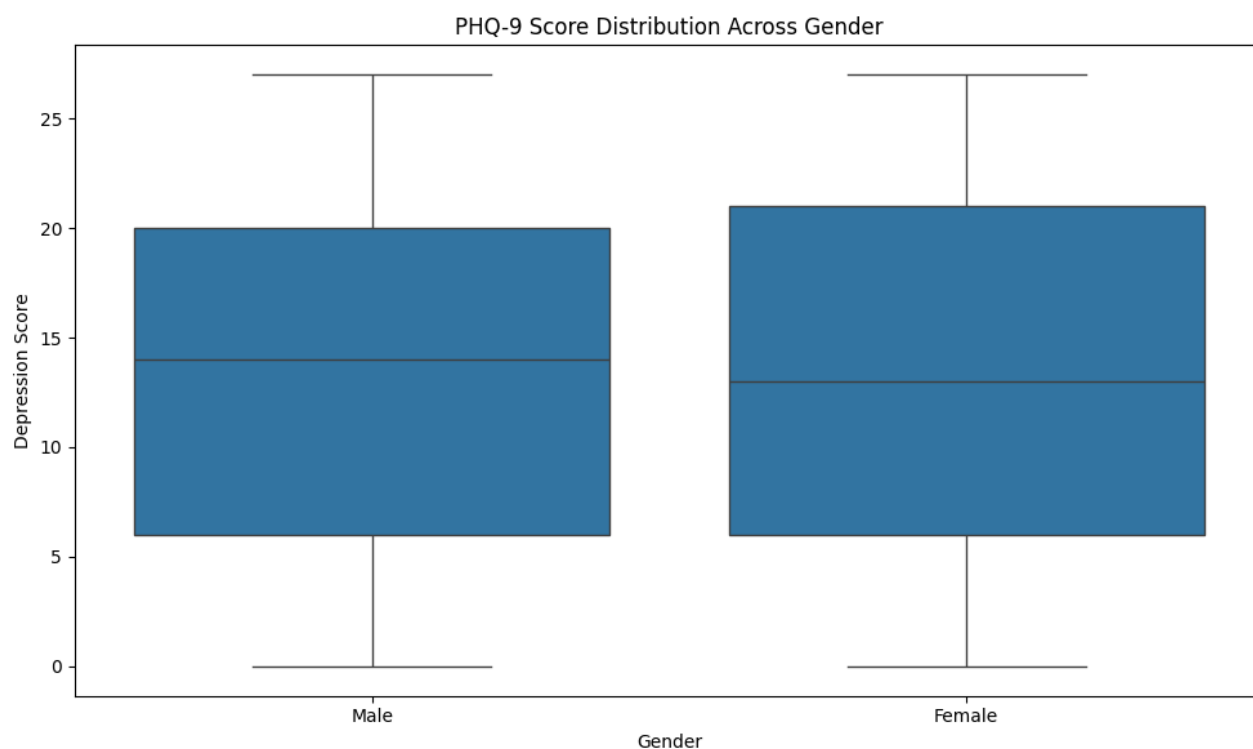
A box plot was generated to compare depression severity distributions across gender groups.

The visualization identified:

- variation in median depression scores,
- differing interquartile ranges,
- and visible outlier values across demographic categories.

Certain groups displayed wider score dispersion, indicating greater variability in depression severity indicators.

The analysis also demonstrated that mental health severity patterns are not uniformly distributed across demographic populations.



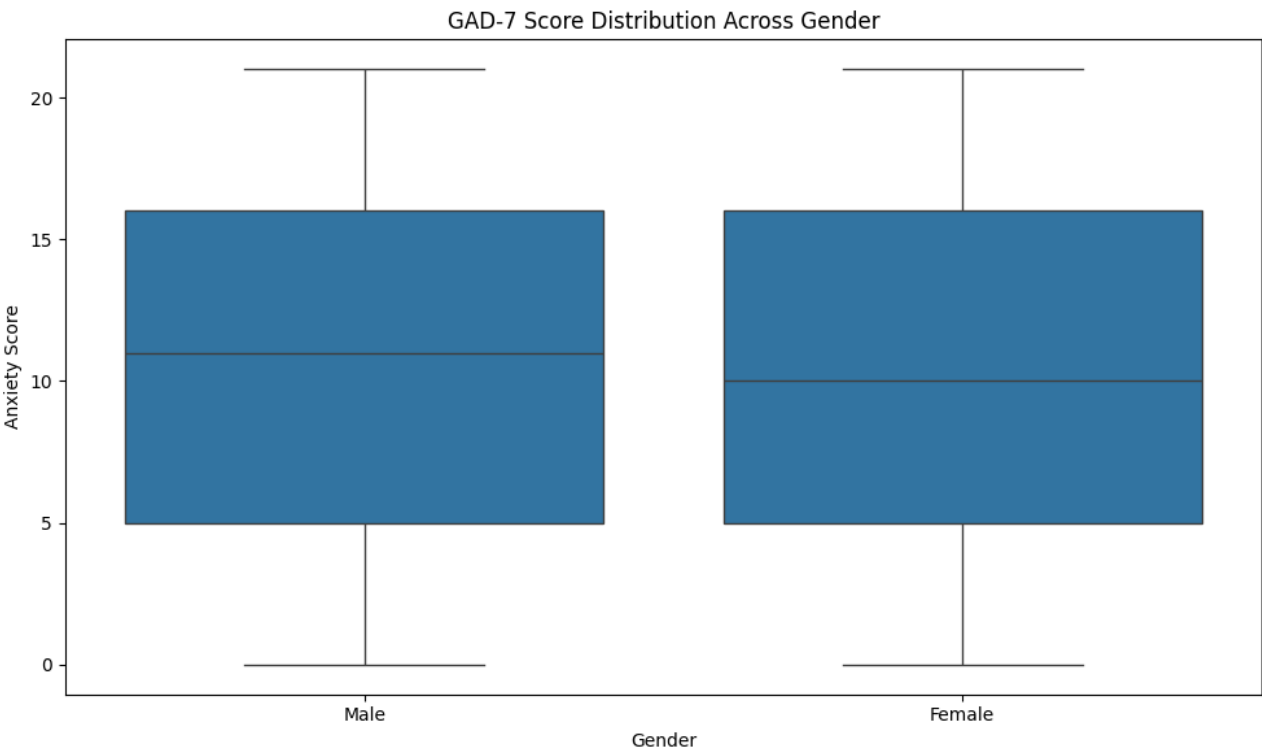
6.2 GAD-7 Equivalent Anxiety Score Analysis:

The anxiety distribution box plot highlighted:

- differences in anxiety score spread,
- varying score concentration,
- and multiple high-score outliers.

The presence of outliers suggests that a subset of participants reported significantly higher anxiety severity levels compared to the overall distribution.

The comparative visualization supported clearer interpretation of demographic anxiety trends.



7. AGE-BASED PHQ/GAD DISTRIBUTION ANALYSIS

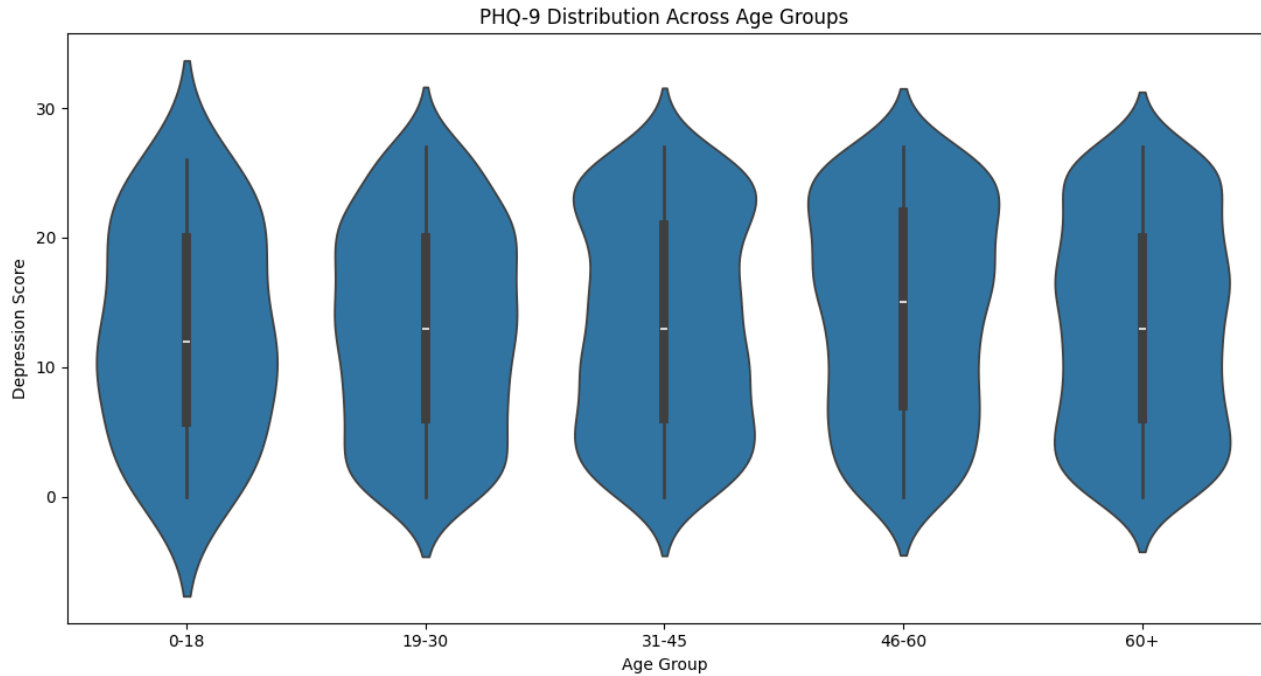
7.1 Depression Score Violin Plot Analysis:

Violin plot analysis was used to examine the density and spread of depression severity scores across age groups.

The visualization demonstrated:

- varying distribution density patterns,
- differences in score concentration,
- and demographic variability in depression severity.

Several age groups showed broader score ranges and denser central distributions, indicating variation in reported mental health severity across the population.

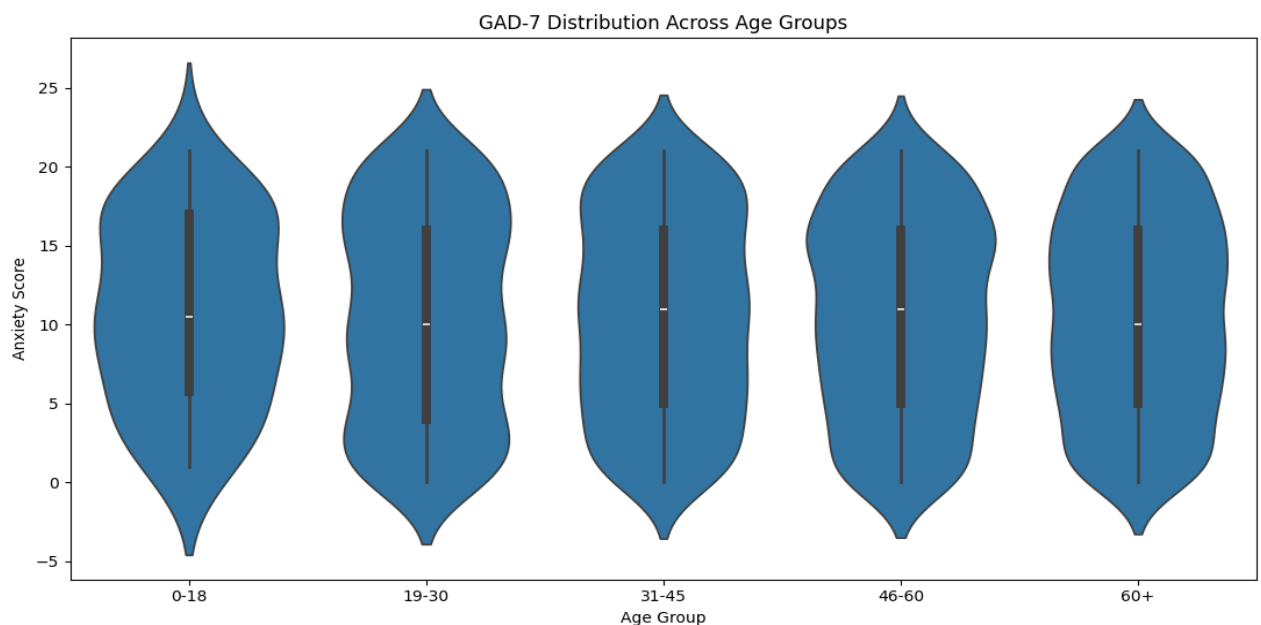


7.2 Anxiety Score Violin Plot Analysis:

The anxiety violin plot demonstrated:

- uneven score distribution across age groups,
- varying density concentration,
- and visible demographic variation in anxiety severity indicators.

Certain age categories showed comparatively wider score distributions, suggesting differing anxiety trends across demographic populations.



8. STATISTICAL OBSERVATIONS

The overall analysis identified:

- moderate distribution skewness,
- visible outlier presence,
- and demographic variability across both depression and anxiety indicators.

The generated distribution charts improved:

- comparative demographic interpretation,
- trend identification,
- and structured exploratory analysis.

The combination of box plots and violin plots provided both:

- summary distribution insight,
- and detailed density visualization.

9. KEY INSIGHTS

- Depression and anxiety severity scores vary across demographic groups.
- Several demographic categories demonstrate wider score variability than others.
- Outlier observations indicate the presence of participants with significantly elevated mental health severity indicators.
- Age-group segmentation improves understanding of population-level distribution patterns.
- Statistical visualizations support clearer exploratory interpretation of mental health datasets.

9. CONCLUSION

The PHQ/GAD score distribution analysis identified visible variation in depression and anxiety severity across demographic groups. The generated box plots and violin plots supported demographic comparison, outlier identification, and distribution analysis using structured mental health survey data. The workflow also demonstrated the usefulness of statistical visualization techniques in exploratory healthcare analytics.